Master Thesis

Configuration-based mobile applications

zur Erlangung des akademischen Grades

Master of Science

eingereicht im Fachbereich Mathematik, Naturwissenschaften und Informatik an der Technische Hochschule Mittelhessen

von

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Erklärung der Selbstständigkeit

Hiermit versichere ich, die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie die Zitate deutlich kenntlich gemacht zu haben.

Gießen, den February 5, 2019 Samuel Schepp

This thesis describes configuration-based mobile applications and their infrastructure. In addition, a possible way of implementing a configuration system is discussed. Finally, there is an evaluation of the use case of interpreting and generated apps.

Furthermore, the implementation is compared to the approach of established literature by classifying the model and interpreter.

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1 Introduction

Mobile application usage is on the rising[Per].

configuration based apps.

how? download layout, data definition, actions, metadata.

all apps use the same container, which is always up to date and uses latest technology.

some metadata has to be baked in on compile time.

1.1 Problem definition

why configuration based?

long waiting times on app store approval

example: customer terminal, city marketing stele: lot of work to redeploy large amount of units.

lot of different apps need same boilerplate code. hard to migrate all to newest technology.

1.2 Motivation

use of native mobile features, camera, background gps, beacons, push notification (with payload). caching, fast ui, good battery life.

no need to depend on third party frameworks, direct use of phone vendor's apis.

1.3 Distinction

This thesis will not describe the development experience nor the initial expense of the project. Neither will this thesis evaluate financial aspects and at which point, the project was or is worth the development costs.

1.4 Structure

1.4.1 Research Question

The thesis aims to answer the question ”Can a code interpreter be integrated in any domain?”. In addition, the development approach is compared to procedu- res, that are described in popular literature.

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2 Background

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2.1 Fabrik19

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2.2 Mobilitysuite

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3 Concept

3.1 Model driven development

3.1.1 Code generators

3.1.2 Code interpreters

3.2 Implementation goals

Same day delivery

easy configuration

simple datasources

common app tasks simplified

content driven (wordpress, social media)

reactive form elements

3.3 Trade offs

boundary conditions

limits to development of app contents (app is out, but the customer needs more functionality, which is not possible: incomprehensible reasons for limits.)

4 Implementation

lorem ipsum

4.1 Components

4.1.1 Nodes

4.1.2 Layouts

4.1.3 Database

4.1.4 Actions

4.1.5 Data proxy

4.2 Infrastructure

4.2.1 Control center

4.2.2 Resource server

4.2.3 Image cruncher

5 Evaluation

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5.1 Analysis

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5.1.1 Benefits

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5.1.2 Disadvantages

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5.2 Classification

5.2.1 Approach

5.2.2 Model

5.3 Conclusion

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Bibliography

[Per] Perkins, Kleiner: Internet Trends Report 2018, URL https://www.kleinerperkins.com/perspectives/internet-trends-report-2018/

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